Robust Regression under Distributional Contamination: A Monte Carlo Study of OLS, LTS, Theil and Bayesian Estimators

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ABSTRACT

The Ordinary Least Squares (OLS) estimator, while standard for linear regression, is highly susceptible to outliers and heavy-tailed error distributions due to its reliance on normality and homoscedasticity. This study utilizes the ADEMP and Tidy Simulation frameworks to conduct an extensive Monte-Carlo simulation (150,000 replications) to evaluate selected estimators within simple linear regression. We performed simulations across six sample sizes and four error-generating environments (Standard, Contamination, Outlier, and Mixture models). Estimators - OLS, Least Trimmed Squares (LTS), Theil's pair-wise median, and Bayesian regression - were compared using bias, mean square error (MSE), Monte Carlo standard error (MCSE), and interval coverage probability. Furthermore, kernel density estimation was employed to assess the empirical behavior, structural stability, and convergence characteristics of the simulated error distributions. Our results demonstrate that Theil's method consistently exhibits the highest reliability across all conditions; however, its computational complexity restricts its use in large-scale datasets. The LTS method offers the optimal balance between computational efficiency and robustness for larger samples. Conversely, standard Bayesian regression, when implemented without robust heavy-tailed priors, remains as vulnerable to outliers as OLS. These findings provide empirical guidance for selecting robust statistical frameworks, confirming that transitioning from OLS is essential when data deviates from Gaussian assumptions.

Keywords: Robust Statistic; Monte Carlo Simulation; Breakdown Point; Estimator Efficiency, Outlier Resistance, Heavy-Tailed Distributions, Simple Linear Regression.

1.0 INTRODUCTION

1.1 Overview of Regression Robustness

Ordinary Least Squares (OLS) is the traditional standard for linear regression due to the Gauss-Markov theorem. However, as noted by Birkes and Dodge (1993), its reliance on normality and homoscedasticity renders it fragile when these assumptions are violated. A primary weakness is its sensitivity to outliers; because OLS minimizes squared residuals, it assigns disproportionate weight to extreme observations, skewing coefficients and invalidating models (Dietz 1987, Rousseeuw and Leroy 1987).

This is formally measured by the breakdown point - the proportion of incorrect observations an estimator can tolerate before failing (Hampel 1974). OLS has a breakdown point of $\frac{1}{n}$ which approaches 0% as sample size increases, leaving it defenseless against influential outliers (Rousseeuw and Yohai 1984). Consequently, robust frameworks like Least Trimmed Squares (LTS) and Theil's pairwise median have emerged as essential alternatives for complex, noisy, or high-dimensional datasets (Morris, White and Crowther 2019, Sami 2023).

1.2 The Rise of Robust Methods

To counter OLS limitations, researchers developed non-parametric and robust estimators. Theil's method uses the median of pairwise slopes, offering rank-invariant resistance to outliers without requiring normal errors (Theil 1950, Hussain and Sprent 1983).

Complementing this, (Rousseeuw and Yohai 1984) introduced LTS, which minimizes the sum of the smallest squared residuals to trim influential observations. While historically computationally demanding, modern MCMC and algorithmic improvements have revitalized these methods, allowing them to outperform classical approaches in heavy-tailed or heteroscedastic environments (Agullo 2001, Ibrahim, Dike and Badawaire 2022).

1.3 Bayesian Integration

Bayesian inference has evolved through MCMC advancements (Bolstad 2010, Gelman et. al. 2013). However, a common misconception is that Bayesian regression is inherently robust. Standard models with conjugate Normal priors mirror OLS vulnerabilities to outliers (Giacomini, Kitangwa and Read 2021). While heavy-tailed likelihoods (e.g., Student-t) can improve robustness, their comparative efficacy against frequentist methods like Theil's remains under-explored. A research gap exists regarding direct, head-to-head simulations of these paradigms under standardized frameworks like ADEMP (Morris, White and Crowther 2019).

1.4 Simulation Methodology

This study evaluates the trade-off between statistical precision and computational efficiency. While Theil's method is highly robust, its $O(n^2)$ complexity limits large-scale application (Theil 1950, Sami 2023). Conversely, LTS offers a superior breakdown point and better scalability. By employing the ADEMP framework and "Tidy Simulation" (van Kesteren 2026)., this research provides a rigorous, reproducible evaluation of these three paradigms - Theil's, LTS, and Bayesian - to resolve debates regarding their performance under non-normal and contaminated data.

2.0 MATERIALS & METHODS

2.1 Simulation Design

This study employs the ADEMP framework (Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures) to conduct a rigorous Monte Carlo simulation. By utilizing the "Tidy Simulation" paradigm, we ensure the experiment is scalable, reproducible, and provides a controlled environment to compare the performance of four regression estimators across varying levels of data contamination.

2.2. Data-Generating Mechanism (DGM)

The simulation considers a simple linear regression model:

$$Y_t = \alpha + \beta X_t + \varepsilon_t \text{ for } \alpha = 5 \text{ and } \beta = 2$$

where the design variable X is generated as a sequential model of the form $X_t = t;\ t = 1,2,3, \dots, n$, and ε_t is the error term. To evaluate estimator performance under non-ideal conditions, we generated errors from three distinct distributions:

1. Normal: $N(0, 1)$ (Baseline).
2. Lognormal: L(0,1) Skewed, non-symmetric distribution.
3. Cauchy: C(0,1) Heavy-tailed distribution with undefined mean and variance.

Furthermore, we introduced three contamination scenarios (see table 1 below): Outliers (extreme values injected into Y - Model_o), Mixtures (a weighted combination of two distributions - Model_m), and Contaminations (data corruption - Model_c). These scenarios were tested across six sample sizes ($n = 10, 30, 50, 100, 500, 1000$) with $N = 150{,}000$ iterations per condition to ensure stability in the performance metrics.

For this study, the alternative error distributions were constructed as:

$$(1 - \epsilon)F(x) + \epsilon G(x)$$

where $F(x)$ represents the standard error distribution, $G(x)$ denotes the contaminating error distribution and ϵ is the probability of contamination.

The empirical behaviors of the simulated error data was characterized using Kernel Density Estimation (KDE) plots across the six sample sizes and under the four distinct structural frameworks considered in the study. Within each model, the three error distributions under investigation (Normal, Lognormal, and Cauchy) were assessed in order to evaluate structural stability, tail behavior, and asymptotic convergence properties under varying degrees of stochastic perturbation.

Table 1: Alternative Error Distribution Models Specification

Distribution	Standard	$Model_o$	$Model_m$	$Model_c$
Normal	N(0,1)	N(0,10)	N(10,10)	G(0.5)
Log-Normal	LN(0,1)	LN(0,10)	LN(10,10)	G(0.5)
Cauchy	C(0,1)	C(0,10)	C(10,10)	G(0.5)

Note: $\epsilon = 0.2$; random seed = sample size (n)

2.3. Mathematical Specification of Estimators

The choice of estimators represents a trade-off between statistical efficiency and robustness. The objective functions for each method are specified below:

- Ordinary Least Squares (OLS): This estimator minimizes the sum of squared residuals:

$$\min_{\beta} \sum_{i=1}^{n} \left(y_i - (\alpha + \beta x_i)\right)^2$$

  OLS provides the Best Linear Unbiased Estimator (BLUE) under Gaussian assumptions but possesses a breakdown point of $\frac{1}{n}$, rendering it highly sensitive to outliers.

- Least Trimmed Squares (LTS): LTS improves robustness by minimizing the sum of the h smallest squared residuals:

$$\min_{\beta} \sum_{i=1}^{h} (r)^2{}_i$$

where $r^2{}_i \leq \cdots \leq r^2{}_n$ are the ordered squared residuals. This “trimming” approach effectively ignores extreme values, allowing for a breakdown point of up to 50% (Rousseeuw, 1984).

- Theil’s Pair-wise Median: This non-parametric estimator calculates the median of all possible slopes between pairs of observations:

$$\hat{\beta} = median\left\{\frac{y_j - y_i}{x_j - x_i}: x_j \neq x_i; 1 \leq i \leq j \leq n.\right\}$$

Theil’s method is rank-invariant and provides high robustness against contamination, regardless of the distribution of the error terms. However, its $O(n^2)$ complexity limits large-scale application (Table 2).

- Bayesian Inference (BAYES): We utilize a Bayesian framework with a Normal likelihood $Y \propto N(X\beta, \sigma^2 I)$. The posterior is proportional to the likelihood multiplied by the prior $p(\beta|y)\ \alpha\, p(y|\beta)$ x $p(\beta)$. We employ Markov Chain Monte Carlo (MCMC) sampling to approximate the posterior. It is noted that with non-informative priors, this model mirrors the vulnerability of OLS to outliers, serving as a control to test the framework’s inherent behaviors in the absence of robust priors.

Table 2: Estimators Computational Time Complexity

Operating System	Windows 10			Time (s)		
GPU	None	**n**	**OLS**	**LTS**	**Theil**	**Bayesian**
Processor	AMD64	10	0.77	0.78	3.01	0.44
Physical Cores	4	30	0.83	2.41	27.13	0.63
Total Cores (Logical)	4	50	0.80	2.07	137.19	0.81
CPU Frequency (Mhz)	1800	100	1.24	3.99	209.38	1.44
Total RAM (GB)	10.90	500	7.68	23.36	3256.41	6.45
Available RAM (GB)	4.99	1000	14.65	48.59	13514.45	25.57

2.4. Performance Measures

To evaluate the estimators, we compute the Bias, Mean Square Error (MSE), and Monte Carlo Standard Error (MCSE). Estimators are ranked using the Relative Mean Square Error (RMSE), defined as:

$$RMSE = \frac{MSE_{ESTIMATOR}}{MSE_{OLSE}}$$

An RMSE < 1 indicates that the robust estimator outperforms the OLS baseline, while an RMSE > 1 indicates poorer performance relative to OLS.

2.5. Empirical Data Analysis

In order to stress test the theoretical findings observed in the simulations, empirical data on Gross Domestic Products (Current GDP US$) and Population (Total) for 266 countries for the year 2018 and 2024 were retrieved from the World Development Indicators (WDI) data website (World Bank 2026).

Regression analysis was conducted on the dataset, and the regression estimators under study were assessed under two conditions: uncontaminated data and data subject to 20% contamination. Performance was evaluated across three distinct sample sizes (n = 10, n = 50, and n = 266). Priors for the intercept (α) and slope parameters (β) were obtained from the OLS estimates of the dataset for the year 2018. A two-sided hypothesis was set up for the regression slope parameter β as follows:

$$H_0 : \beta = 0 \qquad \text{Versus} \qquad H_1 : |\beta| > 0$$

where $\beta \in \{1.289, 1.757, 1.567\}$, the OLS estimates for $n \in \{10, 50, 266\}$ respectively under no contaminations regime.

3.0 RESULTS

3.1.0. Comparative Analysis of Error Distribution Empirical Behavior

3.1.1. Distributional Dynamics under Contaminations

Figures S1 and S2 in appendix A in the Discoveries Supplementary Package illustrate the empirical density functions of the error terms for the Standard and Contaminations error models. Under the standard model, the Normal distribution exhibits classic Gaussian characteristics, with density sharpening around zero as n increases, consistent with the Central Limit Theorem. With 20% contaminations rate, the Normal distribution maintained its Gaussian characteristics for small sample sizes ($n \leq 30$). But as the sample size increases ($n \geq 50$), it becomes severely deformed with multiple localized secondary modes and a depressed peak.

The Lognormal distribution displays early vulnerability under the standard model, producing a secondary "shoulder" even for small sample size ($n = 10$). However, as the sample size n approaches 1000, these effects subside and the distribution converges toward Gaussian-like profiles. With 20% contaminations rate, perturbation is not highly noticeable and the distribution maintains its convergence toward Gaussian-like profiles as the sample size (n) increases.

Across both the standard and contaminations models, the Cauchy distribution exhibits a progressive decrease in peak density as sample size n increases from 10 to 500. Notably, at $n = 500$, the density estimate appears near-flat relative to the Normal and Lognormal distributions. At $n = 1000$, a slight re-concentration of density is observed, although it remains significantly more dispersed than the other distributions.

3.1.2. Response to Outliers and Mixtures

Figures S3 and S4 in appendix A in the Discoveries Supplementary Package illustrate the empirical density functions of the error terms for the Outliers and Mixtures error models. The

Outlier model exposes the fragility of light-tailed distributions. At small n, the Normal distribution manifests a secondary mode - a visualization of the outlier point-mass that the mean fails to absorb. While these tail ripples are eventually subsumed by the primary mode as the sample size (n) increases, the underlying tail remains structurally inflated. The Cauchy distribution, on the other hand, artificially mimics a Gaussian-like shape with a higher peak and narrower base for small sample sizes (n<=30) and then returns back to its true, heavy-tailed profile as sample size (n) approached 1000.

The Mixtures model represents the most severe disruption. At small sample size (n), epistemic uncertainty is pervasive. The Normal and Cauchy distributions exhibit a notable lack of convergence to Gaussian-like features at smaller sample sizes $n \leq 50$, with a progressive reconsolidation of the density peak becoming visually apparent only at $n \geq 100$. As the sample size n increased, two distinct phenomena emerged: (1) the distribution transitioned toward a global Gaussian geometry centred around the theoretical mixture mean; (2) The right tail exhibited distinct, non-smooth 'ripples' characterized by localized secondary modes. Concurrently, the empirical left tail of the Normal distribution demonstrated a rigid truncation anomaly, consistently bounding at a lower threshold of -15 across larger sample iterations. As usual, at $n = 500$, the Cauchy density estimate appears near-flat relative to the Normal distribution.

Across both the Outliers and Mixtures models, the Lognormal distribution consistently presents as a flat line with no discernible density peak across all tested sample sizes ($n = 10$ to $n = 1000$).

3.2.0 Simulation Results of Comparative Estimator Performance

Simulation Results for the Contaminations, Outliers and Mixtures models are reported in Table S1 and S2 in appendix B in the Discoveries Supplementary Package.

3.2.1 Benchmark Performance (The Standard Error Models)

Slope Parameter Estimation: The performance of the estimators varied significantly depending on the underlying error distribution and sample size. For both Normal and Lognormal Distributions, all estimators under study generally demonstrated high stability with minimal bias and MSE across all sample sizes and with increase in sample size. Also, estimates for β were consistently near the true value of 2.0. When subjected to Cauchy-distributed errors, however, performance degraded across all estimators. This was characterized by a sharp increase in MSE, often reaching extremely high values (e.g., 2.1E+05 at n=10 and 3.6E+03 at n=50).

While Coverage Probability remained consistently at 1.0 across most scenarios, the width of the confidence intervals varied drastically. In the presence of Cauchy errors, the intervals became exceptionally wide, reflecting high uncertainty in the parameter estimates compared to the tight, precise intervals observed under Normal or Lognormal conditions. But, as the sample size increased to n=500 and n=1000, all estimators showed improved convergence toward the true parameter value of 2.000 across the tested distributions, though the Cauchy distribution continued to present challenges for MSE stability.

However, in standard Normal conditions, OLSE and Bayesian estimators generally achieved the lowest RMSE values, showing high efficiency. LTS consistently exhibited higher RMSE compared to the others, likely due to the trimming process under non-heavy-tailed data. MCSE values were uniformly low across all methods. Under the Lognormal Distribution, Theils and LTS often outperformed OLSE and Bayesian in terms of RMSE, suggesting a slightly better handling of the skewness inherent in Lognormal data. The Cauchy distribution presented the most extreme results. While all estimators showed massive MSE/RMSE values, LTS and Theils typically demonstrated lower RMSE than OLSE and Bayesian, which suggests better relative robustness to the extreme outliers characteristic of Cauchy errors.

Conversely, OLSE and Bayesian often produced lower MCSE values, indicating higher precision in their highly variable estimates, though this is offset by their larger total error.

Intercept Parameter Estimation: The observed performance of the estimators for the intercept parameter closely mirrors that of the slope parameter. However, Bayesian estimation frequently shows higher bias and lower values in the Normal context. For the Lognormal Distribution, estimators generally show an upward bias with OLSE and Theils consistently overestimating the intercept compared to LTS and Bayesian methods, and the confidence intervals consistently wider than in the Normal case, reflecting increased uncertainty. As sample size increases to n = 1000, estimators under Normal conditions converge tightly. However, under Cauchy conditions, volatility persists, and even large sample sizes do not entirely stabilize the intercept estimates for methods like OLSE and Bayesian.

3.2.2. Impact of Contaminations, Outliers and Mixtures

Slope Parameter Estimation: Under the Contaminations, Outliers and Mixtures models, robustness becomes critical. Even in Normal error situations, Theil and LTS consistently maintained the lowest RMSE and MCSE across all sample sizes, indicating consistent performance. The Bayesian and OLS estimators performed similarly to robust estimators but often exhibit slightly higher RMSE and MCSE. Whenever the error term is non-normal (Lognormal or Cauchy), however, Theil generally ranks first, showing the highest stability and lowest RMSE, consistently keeping the slope near the true value of 2.0. It also maintained the lowest MCSE, which confirms its superior stability even under non-normal error distributions. LTS ranks second, remaining relatively robust but showing higher RMSE than Theil in extreme outlier scenarios. It also occasionally shows extreme instability in very small samples (n=10) under specific contaminant models. The Bayesian and OLS estimators show extremely high, often catastrophic, MSE values (reaching 10^{32} in specific Cauchy model runs), and massive MCSE values (e.g., 10^{18} to 10^{32}) in heavy-tailed settings, and particularly at smaller sample sizes (n=10, 30), which demonstrates significant sensitivity to extreme observations, rendering them unreliable in these conditions.

Intercept Parameter Estimation: The performance of the intercept (α) estimators mirrors the trends observed in the slope parameter, demonstrating that robust methods are essential when dealing with non-ideal data distributions. However, the LTS estimator consistently maintained the lowest RMSE and MCSE across all cells of the simulation and especially with increasing sample size and particularly in Cauchy and outlier-prone models. Though both the Theil and Bayesian estimators show higher RMSE compared to LTS and OLS in the Normal error case, Theil generally ranks second across board, maintaining a high degree of stability even when other methods failed and thus proving to be a stable alternative to non-robust methods. The OLS estimator acts as a reliable benchmark with stable performance only as long as the Normality requirements are satisfied. Otherwise, OLS and Bayesian showed extreme sensitivity, especially to the Lognormal and Outlier models, with MCSE values becoming astronomically high (e.g., 10^{14} to 10^{18}), signifying that these estimators are highly unreliable under these conditions.

3.2.3 Asymptotic Properties and Sample Size

Simulation results confirm the consistency of all estimators, as they show improved stability with a pronounced reduction in bias, MSE and Monte Carlo Standard Error (MCSE) as n increases from 50 to 1000. For small n (50), high volatility is evident, especially in the intercept. Performance divergence between estimators is most pronounced here, where robust methods provide necessary protection against outliers that OLS cannot mitigate. For Large n $(500, 1000)$, estimators reach high stability with lower MCSE and robust coverage. Consequently, the performance gap between OLS and robust methods widens. However, the rate of convergence is much slower for models with heavy-tailed distributions (Cauchy) compared to the Normal case.

3.3. Empirical Data Analysis

The empirical dataset used for this analysis is presented Table S3 and S4 in appendix C in the Discoveries Supplementary Package. Regression analysis was conducted on the empirical dataset, and the performance of the regression estimators under study was evaluated across three distinct sample sizes (n = 10, n = 50, and n = 266).

3.3.1. Empirical Data Analysis Results

Figure S5 and S6 present the scatter plot of the population (total) against GDP before and after contaminating 20% of the observations respectively. Figure S7 and S8 show that the OLS assumptions of normally distributed residuals and homogeneous variance (homoscedasticity) are violated (see appendix C in the Discoveries Supplementary Package). Results of the regression analysis are shown in Table 5 and Table 6 below. It is evident from these tables that the efficacy of the estimators exhibits significant dependence upon both sample size (n) and the presence of data contamination. In uncontaminated datasets, both the Ordinary Least Squares Estimator (OLS) and the Bayesian estimator demonstrate high precision and produce tighter credible intervals across all sample sizes, characterized by lower bias and MSE (approximately 40.277 and 40.283), thereby outperforming Theil (MSE = 50.816) and LTS (MSE = 51.016), particularly at smaller sample sizes (n=10). As the sample size increased to n=266, the performance metrics across all estimators tended to stabilize, with Standard Errors decreasing significantly for all models.

Table 5: Empirical Regression Analysis Results for Data with no Contaminations

	α $\bar{x}$	BIAS	σ	C. Int. (α) LCL	UCL	β $\bar{x}$	BIAS	σ	C. Int. (β) LCL	UCL	MODEL BIAS	MSE
						n=10						
OLS	-0.494	8.893	15.003	-2.271	1.284	1.291	-0.293	0.907	-0.487	3.068	0.000	108.268
Bayes	-1.544	9.943	15.105	-3.333	0.246	1.289	-0.291	0.913	-0.501	3.078	1.086	110.921
Theil	11.177	-2.777	16.196	9.258	13.095	0.771	0.226	0.979	-1.147	2.690	-3.282	136.949
LTS	13.761	-5.361	17.185	11.725	15.796	0.701	0.296	1.039	-1.334	2.737	-4.738	164.505
						n=50						
OLS	-5.435	13.835	5.779	-6.151	-4.720	1.759	-0.762	0.365	1.044	2.475	0.000	49.663
Bayes	-5.670	14.070	5.783	-6.386	-4.954	1.757	-0.760	0.365	1.041	2.473	0.267	49.808
Theil	10.175	-1.775	6.323	9.392	10.958	0.880	0.117	0.399	0.098	1.663	-1.910	63.102
LTS	11.436	-3.036	6.508	10.630	12.241	0.841	0.157	0.411	0.035	1.646	-2.549	69.477
						n=266						
OLS	-2.260	10.659	2.088	-2.507	-2.012	1.568	-0.570	0.127	1.320	1.816	0.000	40.277
Bayes	-2.311	10.711	2.088	-2.559	-2.063	1.567	-0.570	0.127	1.320	1.815	0.056	40.283
Theil	9.503	-1.103	2.263	9.234	9.772	0.957	0.040	0.137	0.688	1.226	-1.863	50.816
LTS	9.477	-1.077	2.265	9.208	9.746	0.961	0.037	0.137	0.692	1.230	-1.896	51.016

The introduction of 20% contamination elucidates the classic bias-variance trade-off within statistical modelling. While OLS remains the most efficient estimator under Gaussian assumptions, its predictive accuracy degrades precipitously in the presence of outliers, particularly at smaller sample sizes. For instance, in a contaminated sample size of $n = 10$, the OLS estimator exhibits a marked escalation in bias compared to its performance within a clean data context. Conversely, The Theil and LTS estimators showed different characteristics under contamination. While OLS and Bayesian remained sensitive to the outliers, the robust estimators exhibited varying degrees of resistance to the contamination, though this often came at the cost of higher MSE compared to the non-contaminated performance. Visual inspection of the scatter plots in figure S5 and S6 further corroborates these findings.

Table 6: Empirical Analysis Results for Data With 20% Contaminations

	α			C. Int. (α)		β			C. Int. (β)		MODEL	
	$\bar{x}$	BIAS	σ	LCL	UCL	$\bar{x}$	BIAS	σ	LCL	UCL	BIAS	MSE
						n=10						
OLS	17.022	-8.622	26.529	13.922	20.122	0.239	0.759	1.582	-2.861	3.339	0.000	138.774
Bayes	15.654	-7.255	26.685	12.536	18.773	0.252	0.745	1.591	-2.866	3.371	1.145	141.723
Theil	14.593	-6.193	28.453	11.268	17.918	0.625	0.372	1.696	-2.700	3.950	-3.982	175.492
LTS	12.191	-3.791	29.868	8.701	15.681	0.847	0.151	1.781	-2.643	4.337	-5.259	203.560
						n=50						
OLS	10.495	-2.095	9.068	9.434	11.556	0.722	0.276	0.541	-0.339	1.783	0.000	126.949
Bayes	10.210	-1.810	9.071	9.149	11.271	0.722	0.275	0.542	-0.339	1.784	0.275	127.103
Theil	11.965	-3.565	9.473	10.856	13.073	0.834	0.164	0.566	-0.275	1.942	-3.318	149.541
LTS	11.656	-3.256	9.520	10.542	12.769	0.864	0.134	0.568	-0.250	1.978	-3.502	152.170
						n=266						
OLS	4.785	3.615	3.911	4.320	5.249	1.010	-0.013	0.237	0.546	1.475	0.000	141.396
Bayes	4.737	3.663	3.912	4.272	5.201	1.010	-0.013	0.237	0.546	1.475	0.048	141.400
Theil	-3.512	11.912	4.054	-3.994	-3.031	1.675	-0.678	0.246	1.194	2.157	-2.488	158.057
LTS	10.043	-1.643	4.065	9.560	10.526	0.891	0.106	0.246	0.409	1.374	-3.327	163.749

The deleterious effects of 20% contamination are starkly evidenced by comparative analysis of the residuals and Normal P-P plots (see Figure S7 – S10 in appendix C in the Discoveries Supplementary Package). In contaminated scenarios, OLS and Bayesian estimators exhibit a pronounced divergence from the reference line (y=x), underscoring their limited capacity to accommodate heavy-tailed or outlier-laden distributions. In contrast, robust estimators demonstrate greater fidelity to the underlying data trend.

The performance of the Least Trimmed Squares (LTS) estimator is particularly salient regarding sample size scalability. In small samples ($n = 10$), LTS displays high variability. However, as the sample size increases to $n = 266$, the robustness of the LTS estimator becomes manifest.

3.3.2. Hypothesis Testing and Estimator Comparison

Looking at the range of the confidence intervals for the contaminated data presented in Table 6 above, constructed for all estimators under consideration at α=0.05, it is observed that, for small sample sizes ($n = 10$), LTS estimators demonstrated the highest precision relative to the null hypothesis by yielding an empirical mean slope (0.847) that is approximately closest to the hypothesized value ($\beta = 1.289$), followed closely by the Theil with an empirical mean slope of 0.625.

The 95% confidence intervals for the Theil's [-2.700 – 3.950] and the LTS [-2.643 – 4.337] estimators both contained the hypothesized value. In contrast, the OLS and Bayesian estimators displayed substantial deviations from the hypothesized parameter in their point estimations, yielding an empirical mean slope of $\beta = 0.239$ and $\beta = 0.252$ respectively, though with a tighter 95% confidence intervals for the OLS [-2.861 – 3.339] and Bayesian [-2.866 – 3.371] estimator respectively.

As the sample size increased ($n = 50$), all the estimators exhibited substantial improvements in precision. However, LTS estimator continued to demonstrate the highest precision, followed closely by the Theil estimator. The 95% confidence intervals for all estimators became significantly tighter and contained the hypothesized values of $\beta = 1.757$ for the sample size. The intervals also indicate that the value of β could potentially be zero and, in some cases, even negative.

When $n = 266$, all confidence intervals became strictly positive with significantly narrower width, effectively excluding zero as a plausible value of β. However, only the confidence intervals for Theil [1.194 – 2.157] contained the hypothesized $\beta = 1.567$. The Theil method now yield the highest precision with an empirical mean slope ($\beta = 1.675$) compared to LTS ($\beta = 0.891$) with confidence interval [0.409 – 1.374]. OLS and Bayes converged to mean slope value ($\beta = 1.010$) with confidence interval [0.546 – 1.475].

4.0 Discussions

The findings of this study underscore the critical trade-off between statistical efficiency and robustness when selecting regression estimators, particularly under departures from the classical Gaussian assumptions. Consistent with theoretical expectations, our simulation and empirical results demonstrate that while traditional methods like Ordinary Least Squares (OLS) and Bayesian estimation provide optimal performance under ideal conditions, their utility is severely compromised in the presence of heavy-tailed distributions and data contamination.

4.1. Evaluation of Estimator Performance

Our analysis reveals that estimator performance is intrinsically linked to the underlying distribution of error terms. In standard Normal conditions, OLS and Bayesian estimators achieve the lowest Root Mean Square Error (RMSE), validating their efficiency when residual normality holds. However, the Cauchy distribution introduces extreme volatility, where OLS and Bayesian methods exhibit catastrophic increases in Mean Square Error (MSE) and Monte Carlo Standard Error (MCSE).

In contrast, robust estimators such as Theil’s slope and Least Trimmed Squares (LTS) demonstrate superior resilience in non-ideal scenarios. Specifically, Theil’s estimator consistently maintains stability and lower RMSE under non-normal error structures, confirming its reliability as a robust alternative. While LTS occasionally displays instability in very small sample sizes ($n=10$), its performance in intercept estimation remains consistently strong, often outperforming alternatives as sample size increases.

4.2. Implications of Contamination and Sample Size

The empirical analysis confirms that the efficacy of these estimators is highly dependent on sample size and the presence of contamination:

- **Sensitivity to Outliers:** The introduction of 20% contamination reveals that OLS and Bayesian estimators are highly sensitive to outliers, with predictive accuracy degrading precipitously, especially in smaller samples.

- **Robustness Scaling:** Although robust estimators (Theil and LTS) experience higher MSE than OLS in uncontaminated settings, they maintain greater fidelity to the underlying data trend under contamination, effectively mitigating the influence of extreme observations.
- **Asymptotic Behavior:** All estimators exhibit improved stability and reduced bias as sample size increases from n=10 to n=1000 (or n=266 in empirical datasets). However, the rate of convergence is significantly slower under heavy-tailed distributions compared to Normal conditions.

4.3. Statistical Inference and Confidence Intervals

The analysis of confidence interval (CI) ranges at the 95% level (σ=0.05) highlights the practical implications of estimator selection for hypothesis testing. Under contaminated conditions and small sample sizes (n=10), LTS and Theil estimators provide point estimates closer to the hypothesized parameters, with CIs that successfully capture these values. While OLS and Bayesian methods produce tighter intervals, their significant deviations in point estimation render them unreliable under these conditions.

At larger sample sizes (n=266), while all estimators exhibit tighter intervals, the Theil estimator demonstrates a unique advantage in precision, frequently yielding intervals that encompass hypothesized values, whereas OLS and Bayesian methods may exclude them due to bias. This suggests that for research applications where data quality cannot be guaranteed, relying solely on non-robust methods poses a substantial risk of inferential error.

4.4. Conclusion

Linear regression traditionally assumes normally distributed errors, under which Ordinary Least Squares (OLS) is optimal. However, real-world data frequently violate these assumptions, necessitating robust alternatives. This study supports the necessity of integrating robust regression techniques when analyzing datasets prone to outliers or heavy tails.

Our findings identify Theil's method as the most robust default compared to other estimators under study. It consistently demonstrated the highest reliability, exhibiting minimal bias and the lowest MSE across most conditions. LTS performed effectively as a secondary alternative, particularly in large-sample scenarios where its superior efficiency-complexity trade-off makes it more scalable than Theil's $O(n^2)$ approach. While Bayesian linear regression proved safer than OLS under approximately normal conditions - offering exact inference conditional on the observed data - it remained sensitive to outliers due to the use of conjugate Normal priors.

In summary, OLS should be reserved for cases where normality is strictly satisfied. For contaminated datasets, Theil's estimator remains the primary candidate for reliability, while LTS provides a robust, efficient solution for large-scale applications.

4.5 Limitations and Future Directions

This research operated under the assumptions of non-stochastic design variables and vertical outliers. Future studies should extend this framework to:

- Stochastic Design Variables: Evaluate performance when predictors themselves are subject to error or contamination.
- Advanced Bayesian Architectures: Employ heavy-tailed likelihoods (e.g., Student-t) and robust, non-conjugate priors to explicitly test their competitiveness against non-parametric methods like Theil's.
- High-Dimensional Complexity: Assess these estimators in models with multiple regressors to determine how robust frameworks scale with increased dimensionality and broader contamination structures.

5.0 List of Supplementary Material

Appendix A: KDE Plots of Error Models across Sample Sizes

Appendix B: Extended Simulation Results

Appendix C: Empirical Data Analysis

Appendix D: Exploratory Data Analysis

6.0. References

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Declarations

Disclosure Statement

The author reports there are no competing interests to declare.

Data Availability Statement

The underlying simulation code developed for this research is publicly accessible GitHub. The repository includes all Python scripts used for the simulation, model estimation, Exploratory Data Analysis and empirical analysis, simulation results tables, and empirical data.